

Supplementary Material for “Manifold Transfer Learning for Multitask Optimization”

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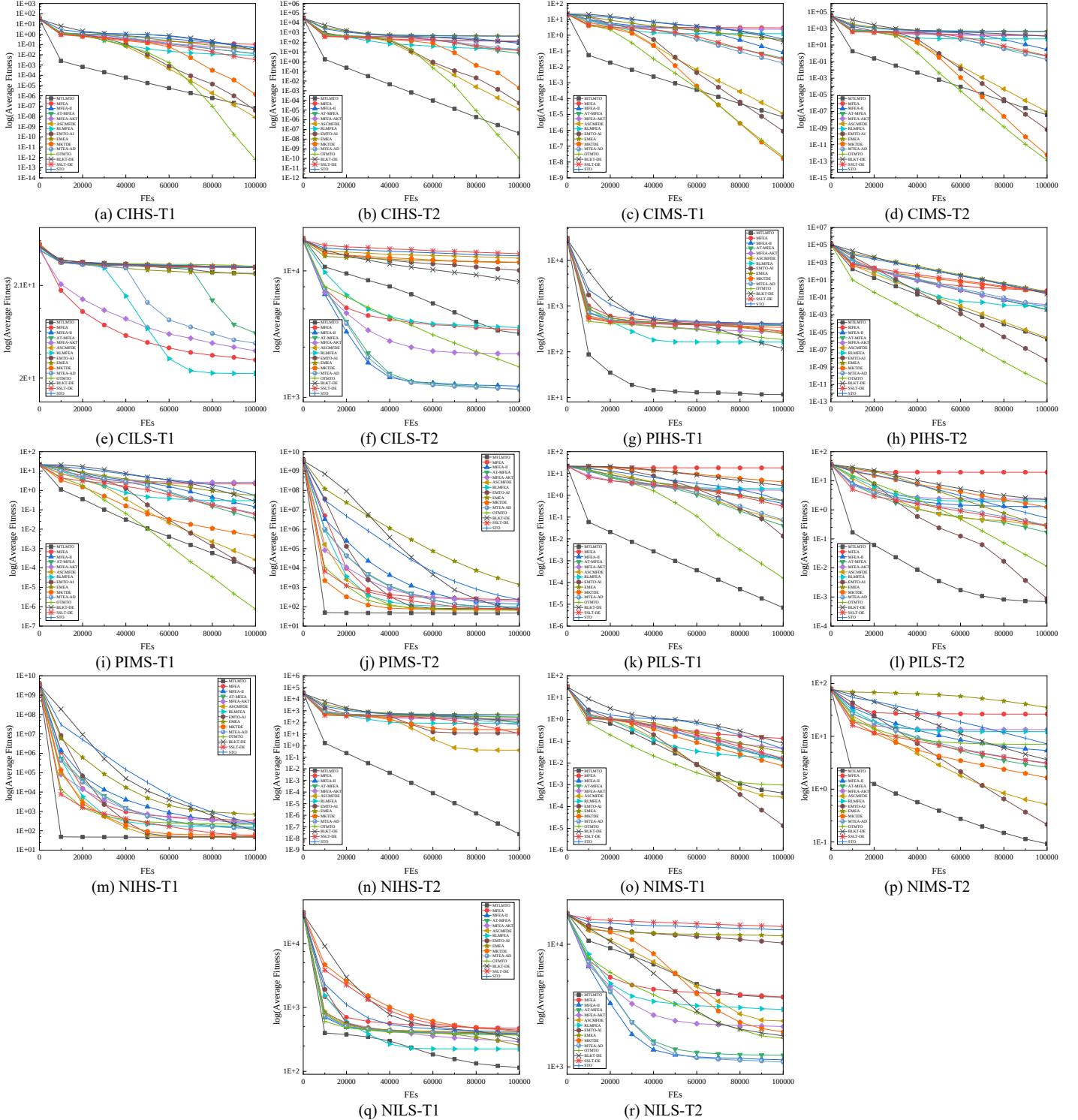


Fig. S1. Convergence curves of different algorithms on the CEC2017 multitask benchmark.

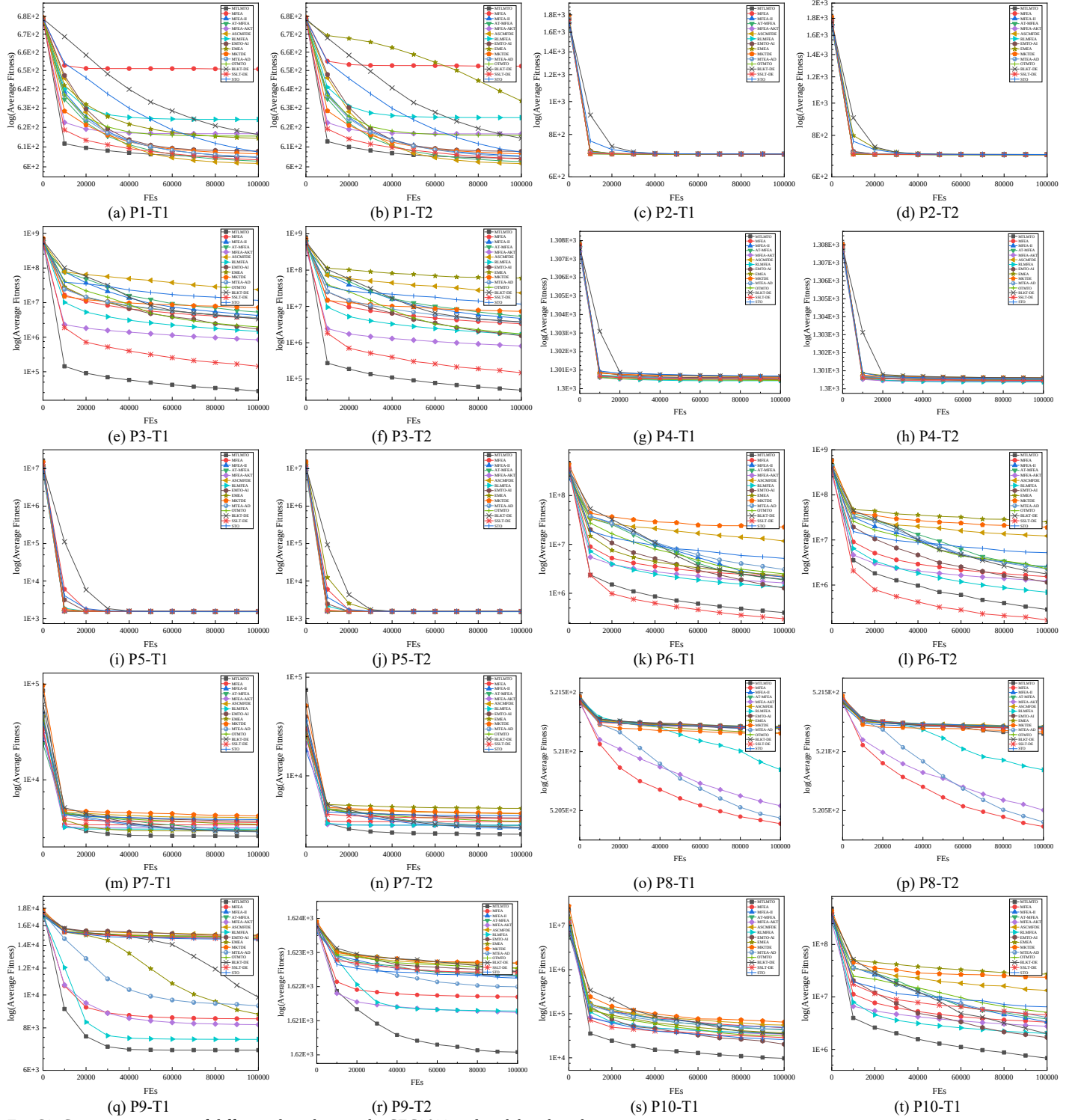


Fig. S2. Convergence curves of different algorithms on the CEC2022 multitask benchmark.

TABLE S.I RESULTS OF MTLMT0 VARIANTS WITH DIFFERENT SETTINGS OF d AND n ON THE CEC2022 MULTITASK BENCHMARK

Problem	MTLMT0	$(d, n) = (1, 3)$	$(d, n) = (1, 7)$	$(d, n) = (1, 9)$	$(d, n) = (2, 3)$	$(d, n) = (2, 5)$	$(d, n) = (2, 7)$	$(d, n) = (2, 9)$
P1-T1	6.05E+02	6.05E+02(≈)	6.05E+02(≈)	6.05E+02(≈)	6.04E+02(−)	6.04E+02(−)	6.04E+02(−)	6.04E+02(−)
P1-T2	6.04E+02	6.04E+02(≈)	6.05E+02(≈)	6.05E+02(≈)	6.03E+02(−)	6.03E+02(−)	6.04E+02(≈)	6.04E+02(≈)
P2-T1	7.00E+02	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)
P2-T2	7.00E+02	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)
P3-T1	2.76E+04	2.57E+04(≈)	2.60E+04(≈)	3.37E+04(≈)	3.90E+04(+)	4.49E+04(+)	4.24E+04(+)	3.24E+04(≈)
P3-T2	4.91E+04	4.27E+04(≈)	4.95E+04(≈)	5.19E+04(≈)	4.78E+04(≈)	6.01E+04(+)	5.82E+04(≈)	5.43E+04(≈)
P4-T1	1.30E+03	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)
P4-T2	1.30E+03	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)
P5-T1	1.53E+03	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)
P5-T2	1.53E+03	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)
P6-T1	4.01E+05	4.88E+05(≈)	4.16E+05(≈)	4.55E+05(≈)	5.45E+05(+)	5.13E+05(≈)	5.82E+05(+)	4.61E+05(≈)
P6-T2	2.87E+05	4.38E+05(+)	4.34E+05(+)	3.19E+05(≈)	3.70E+05(+)	4.06E+05(+)	3.79E+05(+)	4.42E+05(+)
P7-T1	2.60E+03	2.53E+03(≈)	2.65E+03(≈)	2.53E+03(≈)	2.48E+03(≈)	2.53E+03(≈)	2.55E+03(≈)	2.58E+03(≈)
P7-T2	2.56E+03	2.68E+03(≈)	2.64E+03(≈)	2.64E+03(≈)	2.65E+03(≈)	2.59E+03(≈)	2.59E+03(≈)	2.65E+03(≈)
P8-T1	5.21E+02	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)
P8-T2	5.21E+02	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(+)	5.21E+02(+)	5.21E+02(≈)	5.21E+02(+)
P9-T1	6.88E+03	6.94E+03(≈)	7.07E+03(≈)	6.98E+03(≈)	7.13E+03(≈)	6.94E+03(≈)	7.06E+03(≈)	6.99E+03(≈)
P9-T2	1.62E+03	1.62E+03(≈)	1.62E+03(≈)	1.62E+03(≈)	1.62E+03(+)	1.62E+03(≈)	1.62E+03(≈)	1.62E+03(+)
P10-T1	9.72E+03	9.19E+03(≈)	9.98E+03(≈)	9.65E+03(≈)	1.03E+04(≈)	9.32E+03(≈)	1.09E+04(≈)	1.09E+04(≈)
P10-T2	6.81E+05	6.52E+05(≈)	7.33E+05(≈)	7.86E+05(≈)	6.84E+05(≈)	8.23E+05(≈)	7.86E+05(≈)	7.29E+05(≈)
+ / ≈ / −	~	1 / 19 / 0	1 / 19 / 0	0 / 20 / 0	5 / 13 / 2	4 / 14 / 2	2 / 17 / 1	3 / 16 / 1
Problem	$(d, n) = (3, 3)$	$(d, n) = (3, 5)$	$(d, n) = (3, 7)$	$(d, n) = (3, 9)$	$(d, n) = (4, 3)$	$(d, n) = (4, 5)$	$(d, n) = (4, 7)$	$(d, n) = (4, 9)$
P1-T1	6.03E+02(−)	6.04E+02(−)	6.04E+02(−)	6.04E+02(−)	6.03E+02(−)	6.03E+02(−)	6.03E+02(−)	6.03E+02(−)
P1-T2	6.04E+02(−)	6.04E+02(−)	6.04E+02(≈)	6.04E+02(≈)	6.03E+02(−)	6.03E+02(−)	6.03E+02(−)	6.03E+02(−)
P2-T1	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)
P2-T2	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)	7.00E+02(≈)
P3-T1	5.33E+04(+)	5.62E+04(+)	5.65E+04(+)	5.76E+04(+)	8.30E+04(+)	1.15E+05(+)	1.02E+05(+)	7.88E+04(+)
P3-T2	7.85E+04(+)	7.96E+04(+)	8.75E+04(+)	7.98E+04(+)	1.11E+05(+)	1.17E+05(+)	1.23E+05(+)	1.25E+05(+)
P4-T1	1.30E+03(+)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)
P4-T2	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)	1.30E+03(≈)
P5-T1	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)
P5-T2	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)	1.53E+03(≈)
P6-T1	5.59E+05(+)	5.39E+05(+)	5.26E+05(+)	5.16E+05(≈)	5.98E+05(+)	7.31E+05(+)	5.47E+05(+)	6.66E+05(+)
P6-T2	3.89E+05(≈)	4.24E+05(≈)	3.53E+05(≈)	5.04E+05(+)	4.35E+05(+)	4.76E+05(+)	5.18E+05(+)	6.09E+05(+)
P7-T1	2.70E+03(≈)	2.59E+03(≈)	2.54E+03(≈)	2.48E+03(≈)	2.60E+03(≈)	2.50E+03(≈)	2.55E+03(≈)	2.51E+03(≈)
P7-T2	2.61E+03(≈)	2.71E+03(+)	2.60E+03(≈)	2.58E+03(≈)	2.68E+03(+)	2.72E+03(≈)	2.63E+03(≈)	2.71E+03(+)
P8-T1	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(+)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)
P8-T2	5.21E+02(≈)	5.21E+02(+)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(≈)	5.21E+02(+)	5.21E+02(≈)
P9-T1	7.53E+03(≈)	7.60E+03(+)	7.09E+03(≈)	7.25E+03(≈)	8.12E+03(+)	8.07E+03(+)	8.17E+03(+)	8.11E+03(+)
P9-T2	1.62E+03(+)	1.62E+03(+)	1.62E+03(+)	1.62E+03(+)	1.62E+03(+)	1.62E+03(+)	1.62E+03(+)	1.62E+03(+)
P10-T1	9.69E+03(≈)	9.60E+03(≈)	1.14E+04(≈)	1.08E+04(≈)	1.10E+04(≈)	1.15E+04(≈)	1.23E+04(+)	1.01E+04(≈)
P10-T2	9.37E+05(≈)	8.70E+05(+)	1.05E+06(+)	8.43E+05(≈)	7.74E+05(≈)	7.78E+05(≈)	8.77E+05(+)	7.12E+05(≈)
+ / ≈ / −	5 / 13 / 2	8 / 10 / 2	5 / 14 / 1	5 / 14 / 1	7 / 11 / 2	6 / 12 / 2	9 / 9 / 2	7 / 11 / 2